



SQL SUBQUERIES



OVERVIEW

- What is a subquery?
- Subquery Rules
- Subquery with SELECT syntax

WHAT IS A SUBQUERY?

- A subquery is a query within another query and embedded within the WHERE clause
- A subquery is used to return data that will be used in the main query as a condition to further restrict the data to be retrieved.
- A subquery is also known as a nested query or an inner query
- Subqueries can be used with the INSERT, UPDATE, DELETE, and SELECT statements

SUBQUERY RULES

- Subqueries must be enclosed in parentheses
- A subquery may only have one column in the `SELECT` clause, unless multiple columns are in the main query for the subquery to compare its selected items.
- An `ORDER BY` command cannot be used in a subquery, but the main query can use an `ORDER BY`.

SUBQUERY RULES

- Subqueries that return multiple rows can only be used with multiple value operators such as the IN operator.
- The SELECT list cannot include any references to values that evaluate to a BLOB, ARRAY, CLOB, or NCLOB
- A subquery cannot be immediately enclosed in a set function
- The BETWEEN operator cannot be used with a subquery, but it can be used within the subquery.

SUBQUERY WITH SELECT SYNTAX

- Basic SELECT subquery follows this syntax:

```
SELECT column_name [, column_name]
```

```
FROM table1 [,table2]
```

```
WHERE column_name OPERATOR
```

```
(SELECT column_name [, column_name ] FROM table1 [,table2]
```

```
[WHERE])
```